STOCK MARKET ANALYSIS

Six stocks.
One SQL research workflow.

Historical insights and moving-average crossover analysis
01 January 2015 - 31 July 2018

Executive findings

TVS Motors (+86.91%) and Eicher Motors (+82.56%) show the largest increases in supplied raw closes. Bajaj Auto increased 10.05% and Hero Motocorp 6.00%. These compare endpoints in the uploaded files and exclude dividend income and price adjustments.

The data require caution for TCS and Infosys. Their raw changes are -23.82% and -30.91%, respectively. Each series contains a roughly 50% one-session drop. Adjustment factors are absent, so the analysis does not interpret these moves as equivalent shareholder losses.

End-of-sample momentum differs from whole-period change. Bajaj Auto and Infosys have MA20 above MA50 on 31 July 2018. The other four stocks have MA20 below MA50. These are historical regimes, not predictions.

Dataset	Coverage	Analysis
6 source CSVs	889 observations per stock	20/50-session moving averages
5,334 total observations	889 dates shared by all six	114 crossover events

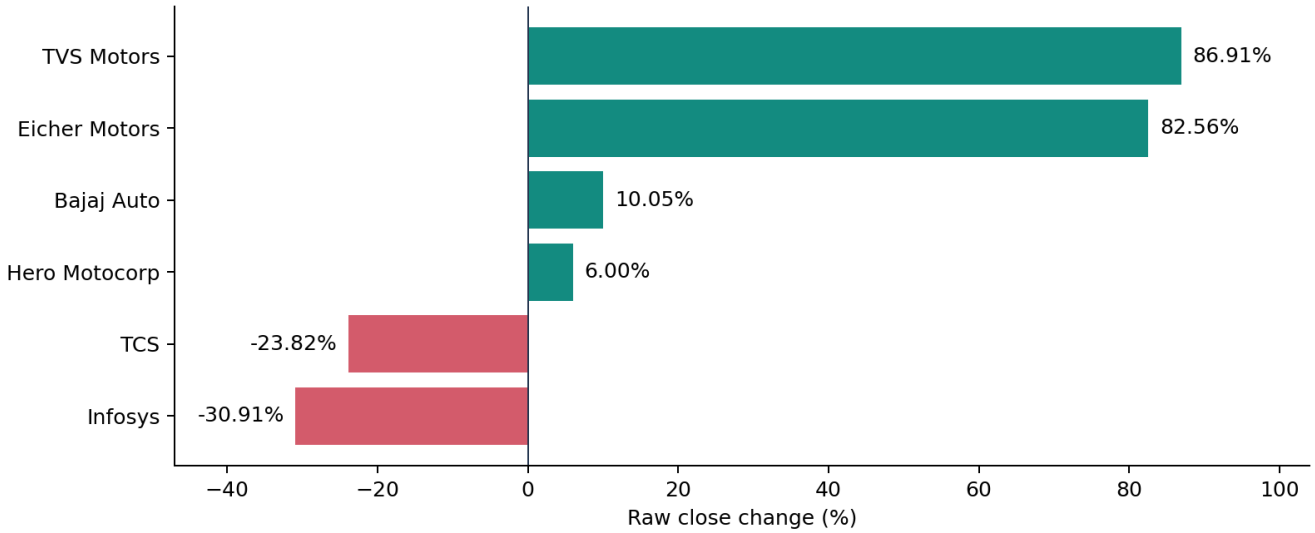
Purpose and scope

Use SQL to compare historical prices, inspect data quality and identify moving-average crossings. A Streamlit interface makes the database searchable through live SQL queries. All results come from the uploaded CSVs; no current prices or external financial claims are used.

Source: uploaded SQL - Stock Market Analysis Project archive, six CSVs and reference PPTX. Prepared 29 September 2026.

01 / Price performance

TVS and Eicher lead raw price growth



Stock	First close (INR)	Last close (INR)	Raw change
TVS Motors	276.85	517.45	+86.91%
Eicher Motors	15,239.15	27,820.95	+82.56%
Bajaj Auto	2,454.10	2,700.70	+10.05%
Hero Motocorp	3,107.30	3,293.80	+6.00%
TCS	2,548.20	1,941.25	-23.82%
Infosys	1,975.80	1,365.00	-30.91%

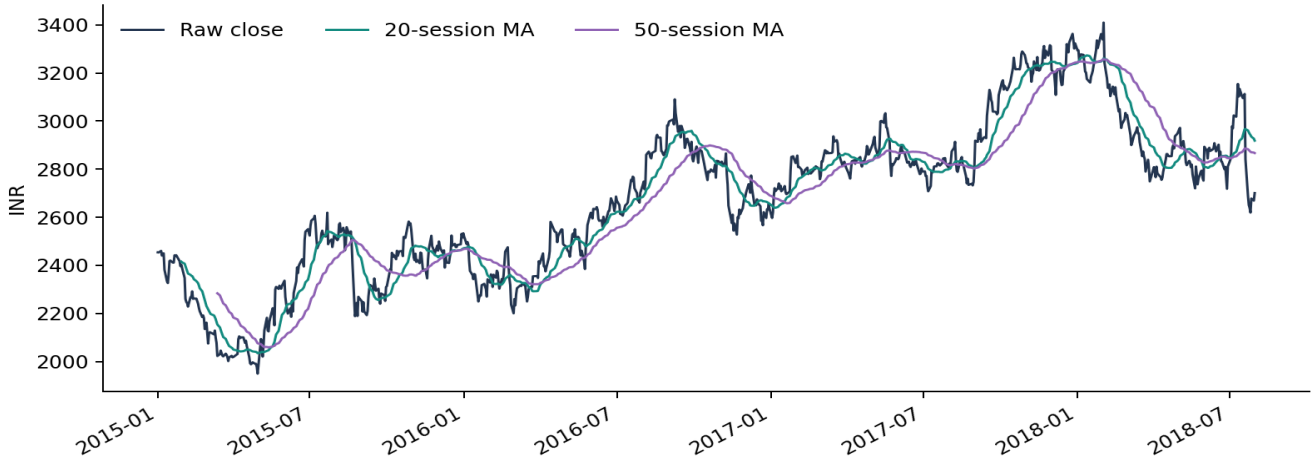
Calculation: 100 x (last observed close / first observed close - 1). Both endpoints are 01 January 2015 and 31 July 2018 for every stock. SQL: stock_summary view (Q7).

Interpretation

The endpoint comparison ranks raw price growth within this supplied sample. It does not measure total return, account for reinvested dividends, or establish that the highest-growth stock is the best investment. Infosys and TCS are especially affected by discontinuities discussed on page 4.

114 crossings across six stocks

Stock	BUY	SELL	31 Jul regime	Last crossing
Bajaj Auto	12	11	ABOVE	2018-06-21 BUY
Eicher Motors	6	7	BELOW	2018-06-06 SELL
Hero Motocorp	9	9	BELOW	2018-05-22 SELL
Infosys	9	9	ABOVE	2018-05-07 BUY
TCS	12	13	BELOW	2018-06-05 SELL
TVS Motors	8	8	BELOW	2018-05-17 SELL



Signal rules

The 20- and 50-session simple moving averages include the current observation and require full windows. BUY occurs when MA20 moves from at or below MA50 to above it. SELL is the reverse. HOLD means no new crossing. The first 50 rows per stock are WARMUP because the first full MA pair lacks a prior full pair.

Bajaj Auto last crossed upward on 21 June 2018; Infosys did so on 07 May 2018. Their MA20 remains above MA50 at the sample end. Signal counts describe events, not completed round-trip trades, trade execution or profits. A signal calculated at the close cannot be assumed executable at that same close.

SQL: moving_averages (Q2), signals (Q3), signal_counts (Q5), latest_signals (Q6). The 20/50-session specification is an explicit project assumption; it reproduces the reference slides' crossover counts.

Two discontinuities limit return interpretation

Stock / date	Prior close	Close	Daily raw change
TCS / 2018-05-31	3,517.75	1,744.80	-50.40%
Infosys / 2015-06-15	1,976.65	991.10	-49.86%

These two observations are flagged by the SQL rule ABS(daily_change_pct) >= 20. The archive contains no adjusted-close series or corporate-action records. The cause is therefore unverified, and no correction factor has been invented.

Stock	Maximum raw drawdown	Average daily turnover (INR million)
TVS Motors	-34.59%	65.30
Eicher Motors	-28.32%	166.44
Bajaj Auto	-23.18%	86.51
Hero Motocorp	-26.90%	116.22
TCS	-52.25%	258.72
Infosys	-62.46%	304.15

Drawdown = 100 x (close / running maximum close - 1); report the minimum. Infosys and TCS drawdowns are distorted by the discontinuities. Turnover describes the supplied file's activity and is not asserted to cover all exchange trading.

Quality checks

All six files have 889 distinct stock/date rows with aligned date coverage. There are no missing or non-positive closes, duplicate stock/date keys, inconsistent OHLC ranges or negative activity values. Deliverable quantity and delivery percentage each have six missing values, one per stock. These remain NULL; SQL AVG excludes NULL rather than treating it as zero.

CSV dates were converted from day-month-name-year text to ISO dates. Rows were sorted chronologically for every window calculation. The original CSVs are preserved. SQL windows and crossover labels were independently checked for all 5,334 observations.

A reproducible SQL project with explicit limits

Corrections to the supplied slides

Reference issue	CSV-based correction
Infosys described as a 3% decline	1,975.80 to 1,365.00 gives -30.91%.
TCS endpoint figures are mixed up	2,548.20 on 01 Jan 2015 to 1,941.25 on 31 Jul 2018 gives -23.82%.
Buy/sell recommendations inferred from signals	Report historical crossover events and regimes. Profitability has not been established.

What the evidence supports

TVS and Eicher had the strongest raw endpoint growth. Bajaj and Infosys were above their 50-session averages on the final date. TCS had the most crossover events (25), followed by Bajaj Auto (23), indicating more crossings under this rule; frequency alone does not establish effectiveness.

What a strategy evaluation would require

Obtain verified adjusted prices and dividend/corporate-action information. Then define next-session execution, transaction costs, slippage, initial holdings and exit rules before backtesting. Compare against an appropriate benchmark and test on a later period. The current archive cannot establish these results.

Reproduce the project

1. Extract Stock_Market_Project.zip and open its folder in VS Code.
2. Run **python -m pip install -r requirements.txt**.
3. Run **python build_database.py** to rebuild the SQL outputs (optional; a populated database is included).
4. Run **python -m streamlit run app.py**.
5. In SQL playground, load an example or enter a SELECT query, then export results.

The SQL dialect is SQLite, not SSMS T-SQL. The app provides live execution on historical data, not live market quotes. The included README documents schema, formulas and validation. A video walkthrough script is supplied because the reference PPTX mentions a video submission.

Traceability

Source data: Bajaj Auto.csv, Eicher Motors.csv, Hero Motocorp.csv, Infosys.csv, TCS.csv, TVS Motors.csv. Brief: SQL- Stock Market Analysis.pptx, slides 2-5. Report tables derive from stock_summary, latest_signals and price_metrics; exported results are in results/. Technical documentation: https://docs.streamlit.io/develop/api-reference.